

# Vedant Singh

B.Tech Student | IIT Palakkad

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## EDUCATION

- **Indian Institute of Technology Palakkad** 2024-Present  
*Bachelors in Technology in Data Science and Engineering; GPA: 8.3* Kerala, India
- **Sunbeam English School** 2024  
*Senior Secondary Education (Class XII): 95.4%* Varanasi, India
- **Sunbeam English School** 2022  
*Secondary Education (Class X): 95.2%* Varanasi, India

## TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, HTML, CSS, SQL
- **AI/ML/Data Science:** NumPy, Pandas, Matplotlib, Seaborn, Langchain (ongoing), Scikit-Learn (ongoing)
- **Web Scraping & Automation:** BeautifulSoup, Selenium
- **Version Control & DevTools:** Git, GitBash

## PROJECTS

- **Document Q/A System using LangChain**  
*Python, LangChain, Vector Embeddings, LLMs*
  - Developing a **document-based question-answering system** using **LangChain** to enable natural language queries over uploaded files.
  - Implemented text ingestion, and chunking for efficient information retrieval.
- **Python Chatbot using Voice Commands and Gemini API**  
*Python, Gemini API, Speech Recognition, Text-to-Speech*
  - Developed a **Python-based voice-enabled chatbot** integrating the **Gemini API** for natural language responses.
  - Implemented **speech-to-text** for voice input and **text-to-speech** for spoken responses.
  - Enabled real-time user interaction through voice commands with seamless **API integration**.
- **Google Maps Review Scraper**  
*Python, Selenium*
  - Built a **Python-based web scraping tool** using **Selenium** to extract ratings and review counts from Google Maps.
  - **Automated** browser interactions using **Selenium** to handle dynamically loaded content.
  - Implemented a terminal-based interface for interactive place queries.
- **Splitzy Pay**  
*PostgreSQL, SQL*
  - Engineered the database architecture using **PostgreSQL** for a scalable expense-sharing platform designed to manage group settlements and track balances.
  - Developed structured **SQL schemas**, authored initial seed data scripts, and optimized common queries to process complex user transaction data efficiently.
  - Designed the relational database model, producing comprehensive **Entity-Relationship (ER) diagrams** and technical specifications for the final project report.
- **Assisted in building key CPU components (MAR, PC, registers) for an 8-bit breadboard computer**
- **IoT-Based Smart Irrigation System**  
*Arduino, Soil Moisture Sensor, IoT, Arduino IoT Cloud*
  - Built an IoT-based soil moisture monitoring system that automatically activates a water pump when moisture levels fall below a threshold.
  - Integrated sensor data with Arduino IoT Cloud to visualize real-time pump status and 7-day moisture trends.
  - Designed an automated control logic to optimize water usage and reduce manual intervention.

## RELEVANT COURSES

- Linear Algebra and Series
- Introduction to Programming
- Introduction to Data Science and Engineering
- Discrete Mathematics
- Introduction to Optimisation
- Data Structures and Algorithms for Data Science
- Probability and Statistics
- Introduction to Artificial Intelligence
- Computer Systems for Data Science
- Database Systems

## POSITIONS OF RESPONSIBILITY

- **Secretary**, Shutterbug - The Photography Club, IIT Palakkad June 2025 - June 2026
- **Core Team Member**, NSS, IIT Palakkad Nov 2025 - June 2026